

Model Description

Notation

Symbol	Description
Sets	
$t \in \mathcal{T}$	Set of timesteps considered in the optimization, $\mathcal{T} = \{0, \dots, T\}$
$n \in \mathcal{N}$	Set of vehicles in the optimization
$d \in \mathcal{D}$	Set of days in the problem, the starts and ends of each day are set such that each day contains one charging transaction per vehicle
$t \in \mathcal{T}_d$	Set of timesteps corresponding to day d
Parameters	
Δt	constant length of each timestep
c_t	Energy price per kWh at time t
b_t	Baseload in kW at time t for the company
E_d^{dem}	The total charging demand on day d
G	Contract capacity of the energy connection in kW
H	Connection capacity of the energy connection in kWh
Q	Total capacity of the battery in kWh
η^+, η^-	Relative loss of energy as a result of charging or discharging the battery respectively
p_t^+, p_t^-	Respectively the maximum and minimum charging speed at timestep t in kW, calculated as the sum of individual charging speed limits for all vehicles present in this timestep.
p_B^*	The maximum charging/discharging speed of the battery
ρ	Cost of energy per kWh if the contracted value, G , is exceeded. Defined as $\rho = 3 \cdot \max_{t \in \mathcal{T}} c_t$
Decision Variables	
P_t^{CP}	Total charging volume in kW in timestep t
ϕ_t	Binary variable: 1 if the pod is charging a vehicle in timestep t , 0 otherwise.
P_t^{B+}	Charging of the battery in timestep t .
P_t^{B-}	Discharging of the battery into one of the vehicles in timestep t . <i>note: negative value indicates discharging of the battery</i>
P_t^{pen}	Charging in kW above the contracted limit in timestep t <i>note: when the contracted capacity is equal to the connection capacity, this variable can be ignored.</i>
Q	Size of the battery, in steps of 100 kWh. <i>note: Q is defined as $100 \cdot q$, where $q \in \mathbb{N}$.</i>
SoC_t	State of charge of the battery, a value between 0 and 1 indicating the fraction of total capacity left at timestep t

Mathematical Formulation

Objective function:

$$\min \sum_{t \in \mathcal{T}} c_t \Delta t (P_t^{CP} + P_t^{B+} - P_t^{B-}) + \rho \Delta t P_t^{pen} \quad (1)$$

Subject to:

$$\sum_{t \in \mathcal{T}_d} \Delta t P_t^{CP} = E_d^{dem} \quad \forall d \in \mathcal{D} \quad (2)$$

$$P_t^{CP} \leq p_t^+ \phi_t \quad \forall t \in \mathcal{T} \quad (3)$$

$$P_t^{CP} \geq p_t^- \phi_t \quad \forall t \in \mathcal{T} \quad (4)$$

$$P_t^{B+} \leq p_B^* \quad \forall t \in \mathcal{T} \quad (5)$$

$$P_t^{B+} \geq 0 \quad \forall t \in \mathcal{T} \quad (6)$$

$$P_t^{B-} \leq p_B^* \quad \forall t \in \mathcal{T} \quad (7)$$

$$P_t^{B-} \geq 0 \quad \forall t \in \mathcal{T} \quad (8)$$

$$SoC_{t+1} = SoC_t + \Delta t \left(P_t^{B+} (1 - \eta^+) - \frac{P_t^{B-}}{(1 - \eta^-)} \right) \quad \forall t \in \mathcal{T} \quad (9)$$

$$SoC_t \geq 0 \quad \forall t \in \mathcal{T} \quad (10)$$

$$SoC_t \leq Q \quad \forall t \in \mathcal{T} \quad (11)$$

$$SoC_0 = SoC_T \quad (12)$$

$$Q \geq 100 \quad (13)$$

$$Q \leq 1000 \quad (14)$$

$$G \geq P_t^{CP} + P_t^{B+} - P_t^{B-} + b_t - P_t^{pen} \quad \forall t \in \mathcal{T} \quad (15)$$

$$P_t^{pen} \geq 0 \quad \forall t \in \mathcal{T} \quad (16)$$

$$P_t^{pen} \leq H - G \quad \forall t \in \mathcal{T} \quad (17)$$

- (1) Minimize the sum of the costs for power that goes into the charge points to charge the vehicles, P_t^{CP} , the power used to charge the battery, P_t^B , and the penalty costs for exceeding the contracted grid capacity, P_t^{pen} .
- (2) The charging demand (E_d^{dem}) of each day, d is met though the total charging volume, P_t^{CP} , in the timesteps corresponding to that day, multiplied by the length of each timestep.
- (3-4) P_t^{CP} is constrained by the minimum and maximum charging power of the vehicle (p_t^- and p_t^+). A minimum charging power is considered in the analysis to account for the fact that many EV models need to be charged continuously above a specific minimum charging power to prevent it from turning to idle mode, at which it does not respond to charging signals anymore.

- (5-8) The battery charge/discharge power at time t , P_t^B , is constrained by the maximum discharge capacity $-p_B^*$ of the battery and the maximum charge capacity p_B^* .
- (9) The State of Charge of the battery at time t , SoC_t , determines how full the battery is at that time t . The state of charge at time $t + 1$ is the sum of SoC_t plus the positive charge sent to the battery, P_t^{B+} . However, because of the battery inefficiency, we lose η^+ of this charge. Similarly, we subtract the negative charge going from the battery into a vehicle, P_t^{B-} . And we also take the efficiency of discharging the battery, η^- , into account here. We multiply by Δt to account for the amount of time the battery is receiving or delivering charge.
- (10-11) The State of Charge is between 0 and the chosen battery capacity Q for each timestep t .
- (12) We force the state of charge of the battery to be the same in the first timestep as in the last timestep.
- (13-14) The battery capacity, Q , can be between 100 kWh and 1000 kWh.
- (15) This constraint makes sure that the charge exceeding the contracted capacity, P_t^{pen} , is always at least the difference between the demanded electricity at timestep t and the contracted capacity, G . If the charging into
- (16) The contracted capacity exceedance must be positive.
- (17) The contracted capacity exceedance cannot be so high that the total energy draw at timestep t exceeds the connection capacity H .
- (16-17) These constraint are dropped in the special case when $G = H$. This can be done because P_t^{pen} will have to be zero for all t in this case.